

Module 2: Frequency, Probability, and Distributions — Answer Key

Prequiz

1. Population: **all students at the college**. Sample: **the 150 students in classes before 9 a.m.** Students who do not take classes before 9 a.m. are not represented by this selection rule. If early-class students differ systematically from other students in work schedules, the sample may not represent the whole college well.

2. Bus:

$$\frac{6}{24} = \boxed{0.25}.$$

Not driving means bus, bike, or walk:

$$\frac{6 + 4 + 2}{24} = \frac{12}{24} = \boxed{0.50}.$$

A **bar chart** is appropriate because transportation method is categorical.

3. A **histogram** is appropriate because the ages are quantitative values grouped into numerical intervals.

4. Known probabilities sum to

$$0.18 + 0.27 + 0.31 = 0.76,$$

so

$$p = 1 - 0.76 = \boxed{0.24}.$$

The complement of the 0.18 event is

$$1 - 0.18 = \boxed{0.82}.$$

- 5.

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6} \approx \boxed{0.1667}.$$

Matching quiz

1. Population: all registered county voters. Sample: the 500 selected voters. The intended strength of the plan is **random selection from the population of interest**, which is designed to give a broader cross-section than a convenience sample from one time or place.

- 2.

$$\frac{18}{60} = \boxed{0.30}.$$

3. **Histogram**, because commute time is numerical and grouped into intervals.

- 4.

$$p = 1 - (0.22 + 0.35 + 0.19) = 1 - 0.76 = \boxed{0.24}.$$

Complement of the 0.35 event:

$$1 - 0.35 = \boxed{0.65}.$$

- 5.

$$P(\text{blue}) = \frac{3}{8} = \boxed{0.375}, \quad P(\text{not blue}) = \frac{5}{8} = \boxed{0.625}.$$